



STRETCHING FUNDAMENTALS

This chapter provides basic background and foundational information that is the core for stretching exercises. Although you can increase joint flexibility without understanding these factors, having a knowledge of stretching fundamentals will enable you to individualize the programs in this book to fit your needs. Joint flexibility, or range of motion, is a function of many components that make up the joint. The resistance to the stretch comes from two general sources: passive structural stiffness and tonic reflex activity. Passive structural stiffness refers to the degree of compliance (or resistance to stretch) found in muscle fascicles, tendons, ligaments, aponeuroses, and joint capsules. An explanation of these structures for a particular joint can be found in the respective chapters. The efforts of the nervous system to maintain muscle tone make up tonic reflex activity. Tonic reflex activity has either peripheral origins (muscle spindles and Golgi tendon organs), central origins (pre- and postsynaptic inhibition), or a combination of the two. An explanation of the involvement of the nervous system can be found in this chapter.

ANATOMY AND PHYSIOLOGY OF STRETCHING

Muscles such as the biceps brachii are complex organs composed of nerves, blood vessels, tendons, fascia, and muscle cells. Nerve cells (neurons) and muscle cells are electrically charged. The resting electrical charge, or resting membrane potential, is negative and is generally around -70 millivolts. Neurons and muscle cells are activated by changing their electrical charges. Electrical signals cannot jump between cells, so neurons communicate with other neurons and with muscle cells by releasing specialized chemicals called *neurotransmitters*. Neurotransmitters work by enabling positive sodium ions to enter the cells and make the resting membrane potential more positive. Once the resting membrane potential reaches a threshold potential (generally -62 millivolts), the cell becomes excited, or active. Activated neurons release

other neurotransmitters to activate other nerves, causing activated muscle cells to contract.

Besides being altered to cause excitation, the membrane potential can be altered to cause either facilitation or inhibition. Facilitation occurs when the resting membrane potential is raised slightly above normal but below the threshold potential. Facilitation increases the likelihood that any succeeding neurotransmitter releases will cause the potential to exceed the threshold. This enhances the chances of the neuron's firing and activating the target. Inhibition occurs when the resting membrane potential is lowered below the normal potential, thereby decreasing the likelihood of reaching the threshold. Usually this prevents the neuron from activating its target.

To perform work, the muscle is subdivided into motor units. The motor unit is the basic functional unit of the muscle. A motor unit consists of one motor (muscle) neuron and all the muscle cells to which it connects, as few as 4 to more than 200. Motor units are then subdivided into individual muscle cells. A single muscle cell is sometimes referred to as a *fiber*. A muscle fiber is a bundle of rodlike structures called *myofibrils* that are surrounded by a network of tubes known as the sarcoplasmic reticulum, or SR. Myofibrils are formed by a series of repeating structures called *sarcomeres*. Sarcomeres are the basic functional contractile units of a muscle.

The three basic parts of a sarcomere are thick filaments, thin filaments, and Z-lines. A sarcomere is defined as the segment between two neighboring Z-lines. The thin filaments are attached to both sides of a Z-line and extend out from the Z-line for less than one-half of the total length of the sarcomere. The thick filaments are anchored in the middle of the sarcomere. Each end of a single thick filament is surrounded by six thin filaments in a helical array. During muscle work (concentric, eccentric, or isometric), the thick filaments control the amount and direction that the thin filaments slide over the thick filaments. In concentric work, the thin filaments slide toward each other. In eccentric work, the thick filaments try to prevent the thin filaments from sliding apart. For isometric work, the filaments do not move. All forms of work are initiated by the release of calcium ions from the SR, which occurs only when the muscle cell's resting membrane potential exceeds the threshold potential. The muscle relaxes and quits working when the calcium ions are restored within the SR.

The initial length of a sarcomere is an important factor in muscle function. The amount of force produced by each sarcomere is influenced by length in a pattern similar in shape to an upside-down letter U. As such, force is reduced when the sarcomere length is either long or short. As the sarcomere lengthens, only the tips of the thick and thin filaments can contact each other, and this reduces the number of force-producing connections between the two filaments. When the sarcomere shortens, the thin filaments start to overlap each other, and this overlap also reduces the number of positive force-producing connections.

Sarcomere length is controlled by proprioceptors, or specialized structures incorporated within the muscle organs, especially within the muscles of the limbs. The proprioceptors are specialized sensors that provide information about joint angle, muscle length, and muscle tension. Information about changes in muscle length is provided by proprioceptors called muscle spindles, and they lie parallel to the muscle cells. The Golgi tendon organs, or GTOs, the other type of proprioceptor, lie in series with the muscle cells. GTOs provide information about changes in muscle tension and indirectly can influence muscle length. The muscle spindle has a fast dynamic component and a slow static component that provide information on the amount and rate of change in length. Fast length changes can trigger a stretch, or myotatic, reflex that attempts to resist the change in muscle length by causing the stretched muscle to contract. Slower stretches allow the muscle spindles to relax and adapt to the new longer length.

When the muscle contracts, it produces tension in the tendon and the GTOs. The GTOs record the change and rate of change in tension. When this tension exceeds a certain threshold, it triggers the lengthening reaction via spinal cord connections to inhibit the muscles from contracting and causing them to relax. Also, muscle contraction can induce reciprocal inhibition, or the relaxation of the opposing muscles. For instance, a hard contraction of the biceps brachii can induce relaxation within the triceps brachii.

The body adapts differently to acute stretching (or short-term stretching) and chronic stretching (or stretching done multiple times during a week). The majority of current research shows that when acute stretches cause a noticeable increase in a joint's range of motion, the person can experience either inhibition of the motor nerves, overlengthening of the muscle sarcomeres, or increased length and compliance of the muscle's tendons. No one is sure of the extent of these changes, but it appears that the muscle shape and cell arrangement, muscle length and contribution to movement, and length of the distal and proximal tendons all play a role. Nevertheless, these transient changes are manifested as decreases in maximal strength, power, and strength endurance. On the other hand, research studies have shown that regular heavy stretching for 10 to 15 minutes three or four days a week (chronic stretching) results in the development of increased strength, power, and strength endurance as well as improved flexibility and mobility. Animal studies suggest that these benefits are due in part to increased numbers of sarcomeres in series.

Likewise, research into stretching for injury prevention has shown differences between acute stretching and chronic stretching. Although acute stretching can help an extremely tight person reduce the incidence of muscle strains, most people appear to gain minimal injury-prevention benefit from acute stretching. People who are inherently more flexible are less prone to exercise-related injuries, and inherent flexibility is increased with heavy stretching three or four days a week. Because of these differences between

acute and chronic stretching, many exercise experts now encourage people to do the majority of their stretching at the end of a workout.

TYPES OF STRETCHES

The stretches featured in this book can be executed in a variety of ways. Most people prefer to do them on their own, but they can also be done with the help of another person. Stretches performed without assistance are referred to as active stretches. Stretches performed with assistance from another person are called passive stretches.

Stretching has come to mean different things to different people, and when doing a simple Internet search, one can discover many stretching techniques. Notwithstanding the various types of stretching promoted by different organizations, there are four basic types of stretches: ballistic, proprioceptive neuromuscular facilitation (PNF), static, and dynamic. All other styles are rooted in these four.

Ballistic Stretching

Ballistic stretches use bouncing movements and do not involve holding the stretch for any length of time. Ballistic stretching can increase range of motion quickly by using the body weight or momentum from each bounce to extend the muscles past their normal range of motion. Because ballistic stretching can activate the stretch reflex, many people have postulated that ballistic stretching has a greater potential to cause muscle or tendon damage, especially in the tightest muscles. However, this assertion is purely speculative, and no published research supports the claim that ballistic stretching can cause injury. Nevertheless, ballistic stretching is not recommended for novices or people with very tight muscles, and its use should be limited to highly conditioned and knowledgeable athletes preparing for strenuous activity.

Proprioceptive Neuromuscular Facilitation Stretching

Proprioceptive neuromuscular facilitation (PNF) stretching refers to a technique that more fully incorporates the actions of the proprioceptors. It usually involves passive stretching combined with isometric muscle contraction either throughout the joint's range of motion or at the end of the range of motion. After moving through the complete range of motion, the muscle is relaxed and rested before it is stretched again. Contracting a fully stretched muscle against resistance relaxes the myotatic reflex and allows a stretch greater than normal. This type of stretching is best done with the assistance of another person. Research has repeatedly shown this technique induces the greatest range of motion, maintains increased range of motion, and increases muscle

strength, especially when done after daily exercise. Most research finds that when performed before exercise, PNF decreases maximal effort performance.

Static Stretching

Static stretching is the most commonly used stretching technique. For most individuals, it is the easiest to perform, and it can readily be done either passively or actively. In static stretching, you extend a particular muscle or group of muscles until you feel increased tension or slight discomfort and then hold the position, usually 15 to 60 seconds. This allows the muscles, fascia, ligaments, and tendons to gradually lengthen, but it can decrease the ability of the nerves to activate the muscle properly. The lengthening of the muscle and joint's connective tissue and the lengthening of the muscle sarcomeres result in a loss of muscle tension, and this coupled with decreased excitability can lead to reduced muscle performance. The duration of the poststretching impairment depends on the amount of stretching done.

Several researchers have questioned the purported benefits of preevent static stretching. Numerous studies have established that preevent static stretching can inhibit almost all components of performance. For instance, preevent static stretching can reduce maximal strength, vertical-jump performance, running speed, and muscular endurance. In addition, research studies have failed to establish a link between preevent static stretching and injury prevention. In fact, a few studies have demonstrated that athletes with high levels of flexibility are more likely to suffer injuries if they stretch before an event than those with moderate flexibility. Some evidence shows that once stretched, extremely tight people are less likely to experience muscle strains. Researchers speculate that this occurs because static stretching reduces the overall strength of the muscle. Strains, pulls, and tears happen when a muscle is forcefully contracted, so reducing the force output makes injury less likely. However, evidence suggests that regular stretching for a minimum of 10 minutes three or four days a week results in increased inherent flexibility, strength, power, and strength endurance as well as improved mobility and maintenance of blood glucose and glycated hemoglobin. Thus, static stretching is most effective after a workout.

Dynamic Stretching

Dynamic stretching is a more functionally oriented stretch that uses sport-specific actions to move limbs through a slightly greater range of motion. Dynamic stretching is generally characterized by swinging, jumping, or other exaggerated movements in which the momentum of the movement carries the limbs up to or slightly past the regular limits of the range of motion. The movements are held for less than 3 seconds. Because the stretch is held for a short time, the muscle is able to lengthen without a reduction in muscle tension or muscle excitability. It also activates a proprioceptive reflex response. The proper acti-

vation of the proprioceptors coupled with the maintenance of muscle tension enables the nerves that activate the muscle cells to fire more quickly, thus enabling the muscle to make faster powerful contractions.

Dynamic stretching is gaining popularity because of the complications that can arise from traditional preevent static stretching. As mentioned, the muscle spindle proprioceptors have a fast dynamic component and a slow static component that provide information about not only the amount of length change, but also the rate of length change. Fast length changes can trigger a stretch, or myotatic, reflex that attempts to resist the change in muscle length by causing the stretched muscle to contract. Slower stretches allow the muscle spindles to relax and adapt to new, longer lengths. Thus, dynamic activities such as running, jumping, and kicking that require quick, forceful movements, use the dynamic receptor to limit flexibility. Consequently, a dynamic stretch that deactivates the dynamic receptor limitation of flexibility can be more beneficial when preparing to perform dynamic activities.

Also, because dynamic stretches increase both muscle temperature and proprioceptive activation, dynamic stretching has been found to be advantageous for improving athletic performance. Dynamic stretching, however, should not be confused with ballistic stretching. Although both involve repeated movements, ballistic movements as explained earlier are rapid, bouncing movements that involve small ranges of movement near the end of the range of motion. Chapter 9 provides several dynamic stretches that can be used as part of an overall stretching program or on their own as needed.

STATIC AND DYNAMIC STRETCHES FOR ATHLETES

Many athletes use static and dynamic stretches in their training programs. Static stretches improve flexibility in certain muscle–joint areas. This type of stretching is the most common approach for improving flexibility. Static stretches hold the stretch of a particular muscle or muscle group for a set time.

Some athletes prefer using dynamic stretches, particularly as part of a warm-up or as preparation for competition. Dynamic stretches stimulate the proprioceptors (stretch receptors), activating their response in an aggressive way by sending feedback to the stretched muscles to be contracted after a quick bouncing motion. Because some athletic events, such as explosive, short-duration activities, could possibly enhance the stimulation of this proprioceptive activation, dynamic stretching prepares athletes better for explosive movements. Such explosive movements might be required to accomplish a certain goal in an athletic event. For example, a person can jump farther and higher if she does a couple of quick up-and-down movements, flexing and extending the hips and knees.

BENEFITS OF A STRETCHING PROGRAM

A regular stretching program can provide several chronic training benefits (see chapter 10 for specific programs):

- Improved flexibility, stamina (muscular endurance), and muscular strength (the degree of benefit depends on how much stress is put on the muscle; chapter 10 explains how this should be done)
- Reduced muscle soreness
- Improved muscle and joint mobility
- More efficient muscular movements and fluidity of motion
- Greater ability to exert maximum force through a wider range of motion
- Prevention of some lower-back problems
- Improved appearance and self-image
- Improved body alignment and posture
- Better warm-up and cool-down in an exercise session
- Improved maintenance of blood glucose

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